

Software Requirements Specification (SRS)

Autonomous Culture & Wellness Intelligence Tool

System Name: PulseIQ

Item	Details
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1. Introduction

1.1 Purpose

The purpose of this Software Requirements Specification (SRS) is to define the functional and non-functional requirements of PulseIQ, an Autonomous Culture & Wellness Intelligence Tool. This document serves as a reference for developers, project supervisors, testers, and stakeholders by describing the system's objectives, features, constraints, and expected behavior throughout the software development lifecycle.

1.2 Project Scope

PulseIQ is a web-based application designed to continuously monitor workplace culture and employee wellbeing through weekly pulse surveys. The platform enables employees to submit regular feedback while providing managers and HR administrators with real-time dashboards, trend analysis, automated notifications, and AI-generated insights. By identifying recurring patterns in employee feedback, the system supports proactive decision-making and timely interventions to improve workplace culture and employee wellbeing.

1.3 Intended Audience

This document is intended for:

- Project Developers
- Project Supervisor
- Project Evaluators
- System Testers
- Future Developers
- HR stakeholders

1.4 Definitions, Acronyms and Abbreviations

Term	Description
AI	Artificial Intelligence
HR	Human Resources
Pulse Survey	A short weekly employee feedback survey
Dashboard	Interface displaying analytics and trends
Employee Wellbeing	Employee's self-reported wellbeing at work
JWT	JSON Web Token
SRS	Software Requirements Specification

1.5 References

- IEEE 830 Software Requirements Specification Guidelines
- React Documentation
- Node.js Documentation
- Express.js Documentation
- MySQL Documentation
- Chart.js Documentation

1.6 Document Overview

This document describes the overall requirements of PulseIQ, including system functions, user roles, operational environment, interfaces, functional requirements, non-functional requirements, and future enhancements.

2. Overall Description

2.1 Product Perspective

PulseIQ is a standalone web-based application developed to improve organizational culture and employee wellbeing through continuous employee feedback collection. Unlike traditional quarterly surveys, the system collects weekly pulse surveys and provides AI-assisted insights, trend detection, and automated notifications.

2.2 Product Functions

The system shall fulfill the following primary functions:

- Allow secure user login.
- Conduct weekly pulse surveys.
- Collect employee wellbeing data.
- Store responses securely.
- Allow optional comments.
- Detect recurring trends.
- Generate notifications.
- Display dashboards.
- Generate AI-powered summaries.
- Produce organizational reports.

2.3 User Classes and Characteristics

- **Employee:** Complete weekly surveys, submit wellbeing feedback, and provide optional comments.
- **Manager:** View team analytics, receive notifications, and access AI-generated summaries.
- **HR Administrator:** Manage users, manage departments, view organization-wide analytics, and receive escalated notifications.

2.4 Operating Environment

The system will operate as a web application using modern web browsers including Google Chrome, Microsoft Edge, Mozilla Firefox, and Safari.

Technology Stack:

- React
- Node.js
- Express.js
- MySQL
- Tailwind CSS
- Chart.js

2.5 Design and Implementation Constraints

- Use free/open-source technologies.
- Secure authentication must be enforced.
- Protect employee privacy.
- AI provides recommendations only and does not make final autonomous decisions.
- The application is developed primarily as a demonstration system.

2.6 Assumptions and Dependencies

- Employees complete weekly surveys honestly.
- Internet connectivity is consistently available.
- Organizations provide complete and accurate employee information.
- AI services remain available.

3. Functional Requirements

ID	Requirement Description	Priority
FR-01	User Authentication: The system shall allow Employees, Managers, and HR Administrators to securely log in using registered credentials.	High
FR-02	Weekly Pulse Survey: The system shall allow employees to complete one weekly pulse survey consisting of: 5 rating questions, optional comments after each question, 1 wellbeing check, and 1 open-ended feedback question.	High
FR-03	Survey Submission: The system shall validate and securely store submitted survey responses.	High
FR-04	Wellbeing Monitoring: The system shall maintain historical wellbeing records for each employee.	High
FR-05	Trend Detection: The system shall detect recurring workplace concerns using predefined rules. Examples include: Low wellbeing for three consecutive weeks, or low manager support ratings for four consecutive weeks.	High
FR-06	Automated Notifications: The system shall notify managers or HR administrators when predefined thresholds are reached.	High
FR-07	Dashboard: The system shall provide dashboards displaying: Survey completion, wellbeing trends, workload trends, team collaboration, manager support, and organizational analytics.	High
FR-08	AI Insight Generation: The system shall generate AI-assisted	Medium

ID	Requirement Description	Priority
	summaries based on employee feedback.	
FR-09	User Management: HR Administrators shall manage: Employees, departments, managers, and user roles.	High
FR-10	Reports: The system shall generate analytical reports for HR and management.	Medium

4. Non-Functional Requirements

- **Performance:** The dashboard must load within three seconds. Survey submission must complete within two seconds.
- **Security:** Enforce password encryption, JWT authentication, and robust role-based authorization.
- **Reliability:** Ensure secure storage of survey responses and strict data integrity during normal operation.
- **Usability:** Provide a clean, user-friendly interface that allows weekly survey completion in under two minutes.
- **Availability:** The application should remain available throughout organizational working hours.
- **Maintainability:** The application shall use a modular architecture to simplify maintenance and future enhancements.

5. External Interface Requirements

- **User Interface:** The system shall provide a Login Page, Employee Dashboard, Weekly Pulse Survey, Manager Dashboard, HR Dashboard, and Notification Page.
- **Software Interface:** React, Express.js, Node.js, MySQL, and relevant AI APIs.
- **Hardware Interface:** Desktop or laptop computers equipped with standard internet connectivity.

6. User Roles and Permissions

User Role	Permissions
Employee	Submit surveys, submit wellbeing responses, and add optional comments.
Manager	View team dashboard, receive automated notifications, and access AI-generated insights.
HR Administrator	Manage users, manage departments, access comprehensive dashboards, generate reports, and manage notifications.

7. Future Enhancements

Potential future improvements to be integrated into later iterations include:

- Mobile application development
- Direct email notifications
- Microsoft Teams integration
- Slack workspace integration
- Machine Learning trend prediction models
- Personalized employee wellbeing recommendations
- Inter-department benchmarking
- Anonymous organization-wide comparative metrics